

Foundations of Cybersecurity/Applied Cryptography

04 July 2023

EXERCISE N. 1 [ALL]

[12]

1. Shannon's definition of perfect cipher and the related intuitive explanation.
2. Statement and proof of the Shannon's Theorem.

EXERCISE N. 2 [ALL]

[12]

Alice and Bob share a secret password P. Design a key establishment protocol that satisfies perfect forward secrecy and direct authentication.

EXERCISE N. 3 [2022-23]

[6]

For each of the following statements, specify whether it is true or not. Argue the answer.

1. A Vernan cipher is unconditionally secure provided the key is at least 128 bits.
2. In a computationally secure cipher, nowadays the key must be at least 80 bits.
3. Let E be a block cipher with $n = 128$ and $k = 64$, then E may be unconditionally secure.
4. Let E be a block cipher with $n = 128$ and $k = 64$, then E may be computationally secure.
5. An unconditionally secure cipher is also computationally secure.

EXERCISE N. 3 [EARLIER]

[6]

Find, explain, and patch the vulnerabilities, if any, in the following function.

```
unsigned int do_stuff(unsigned int a, unsigned int b){
    if(a*b > UINT_MAX){
        /*Handle Error*/
    }
    return a*b;
}
```

Solution

EXERCISE N.1

See theory.

EXERCISE N.2

M1 $A \rightarrow B:$ A, B, n_a
M2 $B \rightarrow A:$ $\{B, A, n_a, n_b, tpubk_b\}_{k_P}$
M3 $A \rightarrow B:$ $\{A, B, n_b, k_s\}_{tpubk_b}$
M4 $B \rightarrow A:$ $\{B, A\}_{k_s}$

where k_P is a key derived from the password P . The key derivation function may be a hash function, for example. The key $tpubk_b$ is the public component of an ephemeral public-private key pair that Bob newly generates for every execution of the protocol (in case we choose to use EDH, it includes p and g too). The pair is securely deleted after message M3. The k_s is the session key generated by Alice. Message M4 is for direct authentication.

If Alice or B is compromised, the long-term secret password P gets compromised as well. The adversary may thus obtain k_P and then $tpubk_b$. However (s)he cannot decrypt M3 and obtain k_s .

EXERCISE N.3 [2022-23]

1. *A Vernan cipher is unconditionally secure provided the key is at least 128 bits.* **FALSE.** Vernam cipher is unconditionally secure if the key is perfectly random.
2. *In a computationally secure cipher, nowadays the key must be at least 80 bits.* **TRUE.**
3. *Let E be a block cipher with $n = 128$ and $k = 64$, then E may be unconditionally secure.* **FALSE.** A perfect block cipher implements all the possible permutations whose number is $P = 2^n! = 2^{128}!$ which is in the order of $n \times 2^n$. Being the key 64 bit, the number of permutations that the cipher actually implements is $K = 2^k = 2^{64}$ which is much smaller than P .
4. *Let E be a block cipher with $n = 128$ and $k = 64$, then E may be computationally secure.* **FALSE.** The key must be at least 80 bit nowadays.
5. *An unconditionally secure cipher is also computationally secure.* **TRUE.**

EXERCISE N.3 [EARLIER]

The problem is wraparound. The error condition is poorly written since if $a*b$ will wrap-around, the if-clause will never be true. The following patch must be applied.

```
unsigned int do_stuff(unsigned int a, unsigned int b){
    if(b!=0 && a > UINT_MAX/b){
        /*Handle Error*/
    }
    return a*b;
}
```

WARNING! The if-clause cannot be written in the inverse order because $(a > \text{UINT_MAX}/b)$ WILL raise an error if b is zero!